
Fixed Income Analysis Report

Bond Futures and the Cheapest-to-Deliver Option

Forward Carry, Conversion Factors,
Implied Repo and Scenario Analysis

This report solves the three written questions in Assignment 3. It prices the three-month forward on the specified Treasury, computes the exchange conversion factor and normalized forward price, recovers the implied repo rate from the quoted futures price, identifies the cheapest-to-deliver selection rule, and sets out a practical framework for forecasting CTD switches.

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Course FRE-GY-9743

Source problem Jianing (Jay) Yao, Assignment 3
September 28, 2026

Abstract

This report analyses the pricing and delivery economics of a Treasury bond futures contract. It derives the three-month forward value from cash-and-carry, converts the bond to the exchange's standard 6% yield basis, and obtains the normalized forward price. It then recovers the implied repo rate from the quoted futures price and states the cheapest-to-deliver rule. Finally, it develops a scenario procedure for forecasting CTD switches as yields, coupon carry, accrued interest, financing conditions and time to delivery change. Price units, accrued-interest treatment and financing conventions are stated explicitly throughout.

Keywords: Bond futures • Conversion factor • Implied repo • Cheapest to deliver • Net basis

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Notation and Modelling Conventions

All prices are per \$100 face amount. The bond pays a 4% annual coupon in two equal installments, so each semiannual coupon is

$$C = \frac{4}{2} = 2.$$

The statement that the bond has *exactly* nine years remaining is taken to place the valuation date on a coupon date. Hence the spot accrued interest is zero. The delivery date is three months later, halfway through the next six-month coupon period, and no coupon is paid between valuation and delivery. The repo quote is treated as a simple annual rate with $T = 3/12 = 0.25$.

Table 1. Inputs and price conventions

Item	Value or convention
Spot clean price	$S_{0,\text{clean}} = 100$
Spot accrued interest	$AI_0 = 0$, so $S_{0,\text{dirty}} = 100$
Coupon and frequency	4% annually; $C = 2$ every six months
Remaining maturity	9 years, or 18 semiannual periods
Forward tenor	$T = 0.25$ year
Market repo rate	$r_{\text{mkt}} = 2\%$ simple annual
Conversion-factor yield	6% nominal, compounded semiannually
Delivery accrued interest	$AI_T = 2(3/6) = 1$

Key result. The accrued-interest adjustment is essential. The financed spot position grows to a dirty value of 100.50, while the comparable quoted forward price is clean and therefore equals 99.50.

1 Q1: Normalized forward price

Question focus. Compute the three-month forward price, the conversion factor, and the normalized forward price; then explain what the conversion factor accomplishes.

1.1 Three-month forward price

Because no coupon is paid before delivery, cash-and-carry gives the dirty delivery value

$$B_F^{\text{dirty}} = S_{0,\text{dirty}}(1 + r_{\text{mkt}}T) = 100(1 + 0.02 \times 0.25) = 100.50. \quad (1.1)$$

At delivery, three months of the next semiannual coupon have accrued:

$$AI_T = C \frac{3}{6} = 2 \times \frac{1}{2} = 1.00. \quad (1.2)$$

The clean forward price used with the conversion factor is therefore

$$B_F = B_F^{\text{dirty}} - AI_T = 100.50 - 1.00 = 99.50. \quad (1.3)$$

1.2 Conversion factor

At the exchange's 6% standard yield, the semiannual yield is 3%. With 18 remaining coupon periods, the clean model price is

$$P_{6\%} = \sum_{i=1}^{18} \frac{2}{(1.03)^i} + \frac{100}{(1.03)^{18}} \quad (1.4)$$

$$= 86.2464869. \quad (1.5)$$

The conversion factor is this clean price divided by 100:

$$K = \frac{86.2464869}{100} = 0.8624648692. \quad (1.6)$$

1.3 Normalized price and interpretation

The normalized forward price is

$$\frac{B_F}{K} = \frac{99.50}{0.8624648692} = 115.3670179 \approx 115.3670. \quad (1.7)$$

The conversion factor maps bonds with different coupons and maturities onto a common 6% yield benchmark. A low-coupon bond such as this 4% issue has $K < 1$, so its clean delivery value is divided by a number below one. This normalization makes a basket deliverable against one futures contract, but it is not perfect: the actual yield curve, coupon carry, repo specialness and accrued interest differ across bonds. Those residual differences create the cheapest-to-deliver option. Accrued interest is excluded from both clean bond prices and the conversion factor and is added separately to the delivery invoice.

2 Q2: Cheapest-to-deliver and implied repo

Question focus. Use the quoted futures price to recover the bond's implied repo and state the economic rule for selecting the CTD bond.

2.1 Implied repo for the specified bond

Let the quoted futures price be $F = 115.3670$. The clean invoice component is FK and the dirty amount received upon delivery is $FK + Al_T$. Since there is no intervening coupon, the implied simple repo rate solves

$$S_{0,\text{dirty}}(1 + r_{\text{imp}}T) = FK + Al_T. \quad (2.1)$$

Hence

$$r_{\text{imp}} = \frac{1}{T} \left(\frac{FK + Al_T}{S_{0,\text{dirty}}} - 1 \right) \quad (2.2)$$

$$= \frac{1}{0.25} \left(\frac{115.3670(0.8624648692) + 1}{100} - 1 \right) \quad (2.3)$$

$$= 0.01999938. \quad (2.4)$$

Therefore

$$r_{\text{imp}} \approx 1.99994\% \approx 2.00\%. \quad (2.5)$$

The small difference from exactly 2% is caused only by rounding the futures quote to four decimal places. This confirms the consistency between Questions 1 and 2.

2.2 How to select the CTD bond

For each deliverable bond x , calculate its implied repo rate from the observed futures price, its conversion factor and all relevant carry cash flows. A trader can buy the bond, finance it in repo and deliver it into the short futures position. If its implied repo exceeds its actual financing rate, this trade has positive carry before costs.

When all bonds can be financed near the same general-collateral rate, the CTD is the bond with the **highest implied repo rate**. Equivalently, it has the lowest normalized forward delivery cost:

$$\text{CTD} = \arg \min_{x \in \mathcal{X}} \frac{B_{F,x}}{K_x} = \arg \max_{x \in \mathcal{X}} r_x^{\text{imp}}. \quad (2.6)$$

If individual bonds trade at different repo rates, especially when one is “special,” the comparison should use financing-adjusted carry. In practice, the preferred bond maximizes the implied financing advantage $r_x^{\text{imp}} - r_x^{\text{actual}}$, or equivalently minimizes net basis after coupon, financing and transaction-cost adjustments.

Key result. For the bond in Question 1, the quoted futures price 115.3670 produces an implied repo of approximately 2%, exactly matching the assumed market repo rate. Its cash-and-carry trade is therefore at break-even before costs.

3 Q3: Scenario analysis and CTD switches

Question focus. Predict which bond may become CTD at a future horizon and explain why the delivery choice matters.

3.1 A reproducible scenario procedure

At a horizon $h = N/365$, the CTD forecast should be recomputed rather than held fixed. For every bond x in the delivery basket:

1. Project its clean price $S_x(h)$ under the scenario yield curve. Roll the coupon schedule forward and compute the accrued interest at the horizon.
2. Apply the scenario repo rate for that specific bond. Include coupons paid between h and delivery and their reinvestment or financing treatment.
3. Compute the scenario forward delivery price $B_{F,x}(h)$ and divide by the exchange-fixed conversion factor K_x .
4. Select the minimum normalized forward price. As a cross-check, use the observed scenario futures price to compute implied repo for every bond and select the largest financing advantage over actual repo.

Formally, the scenario CTD is

$$x^*(h) = \arg \min_{x \in \mathcal{X}} \frac{B_{F,x}(S_x(h), r_x(h), T_d - h)}{K_x}. \quad (3.1)$$

3.2 What can cause a switch?

Table 2. Main drivers of CTD changes

Driver	Effect on the ranking
Yield level and curve shape	Bonds have different duration and convexity, so a parallel move or curve twist changes their clean prices by different amounts.
Coupon carry and accrued interest	Coupon dates, accrued interest and reinvestment alter the cost of carrying each bond to delivery.
Repo specialness	A bond that is expensive to borrow or unusually cheap to finance can move sharply in implied-repo and net-basis rankings.
Passage of time	Remaining financing periods shorten and coupon entitlements change as the contract approaches delivery.
Conversion-factor bias	Conversion factors are fixed from a 6% benchmark and do not update with the actual yield curve, leaving residual relative-value exposure.

CTD identity matters because the futures price is anchored by the cheapest delivery route. The CTD determines the relevant cash bond, conversion factor, duration and DV01 used in a hedge. A CTD switch can therefore change hedge ratios and produce basis P&L even when the broad level of rates moves as expected. For a short futures position it also determines the bond that should be acquired and delivered; for a long position it describes the quality option granted to the short.

Key result. A CTD forecast is a relative-value calculation across the entire basket. Reprice every candidate under the same market scenario, include bond-specific financing, and choose the minimum normalized delivery cost rather than extrapolating today's CTD mechanically.

Summary of the Three Answers

Table 3. Numerical results and decision rules

Quantity	Answer
Three-month dirty forward value	100.5000
Delivery accrued interest	1.0000
Clean forward price B_F	99.5000
Conversion factor K	0.8624648692
Normalized forward price B_F/K	$115.3670179 \approx 115.3670$
Implied repo at $F = 115.3670$	$1.99994\% \approx 2.00\%$
CTD rule under common GC financing	Highest implied repo, equivalently lowest normalized forward cost
CTD rule with bond-specific repo	Highest financing-adjusted implied-repo advantage, equivalently lowest net basis